LetsMeet SQL Case Study

Project overview

LetsMeet is a fictional platform that helps people find local groups and attend events in New York, Chicago, and San Francisco. The project reviews its six-table relational database to understand how Sales, Engineering, and leadership could report on membership, event activity, and venue use.

Approach

I reviewed the table definitions and sample records, identified what each row represents, and traced how groups connect memberships, events, categories, and cities. I then identified reporting questions the schema supports and checks needed before comparing results.

Findings from the schema review

Membership measures. Each membership row links one member to one group. Someone who joins several groups appears several times, so counting membership rows measures participation across groups, while counting distinct member IDs measures people.

Event activity by interest and location. Events link to their organizing groups and venues. The group's category and city fields support comparisons by topic and group location. The venue's address and coordinates identify where an event takes place.

Interest and attendance. The event table stores yes and maybe RSVPs, waitlist counts, RSVP limits, and headcount. Those fields could support reporting on demand and capacity once their definitions and completeness are checked.

Geographic data quality. The case study flags the city table for cleaning. Checking city identifiers and group-to-city matches would make comparisons across markets more reliable.

Scope and next steps

The case-study slides provide schema definitions and sample rows. The findings identify what the database can support. Membership totals, attendance rates, and city rankings would require queries against the full dataset.

The original case study and database schema follow.



LetsMeet

Connecting real people in real life.

What is LetsMeet?

LetsMeet is an online network for professionals and enthusiasts to meet up in real life with groups that share common interests.



Discuss your favorite books together



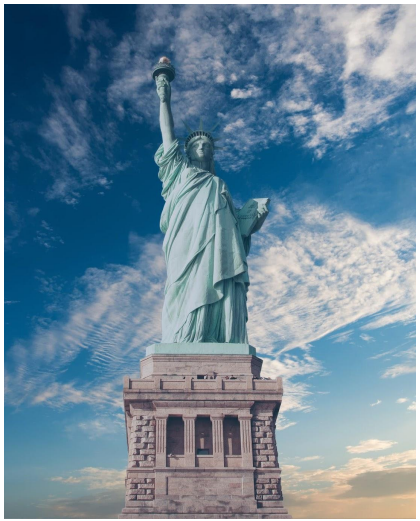
Join a human-centered-design learning event



Sketch in the gallery at the museum

Where is LetsMeet?

We are located in **New York**, **Chicago**, and **San Francisco**.



New York



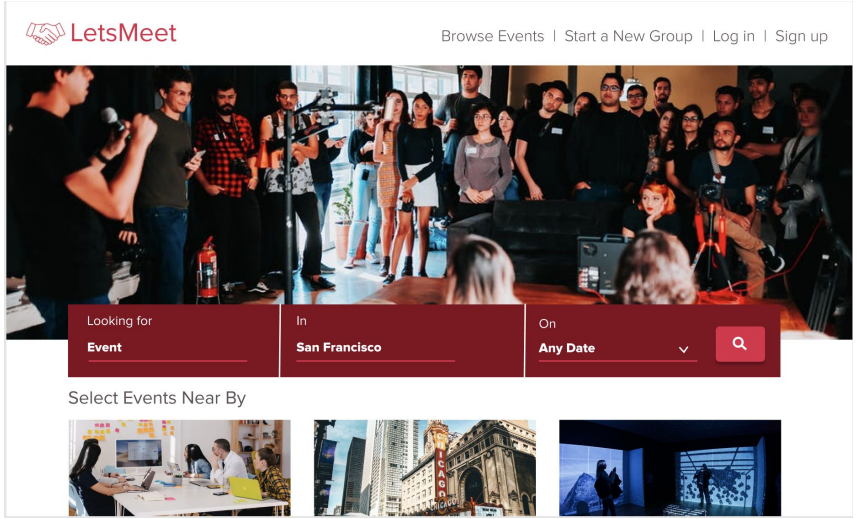
Chicago

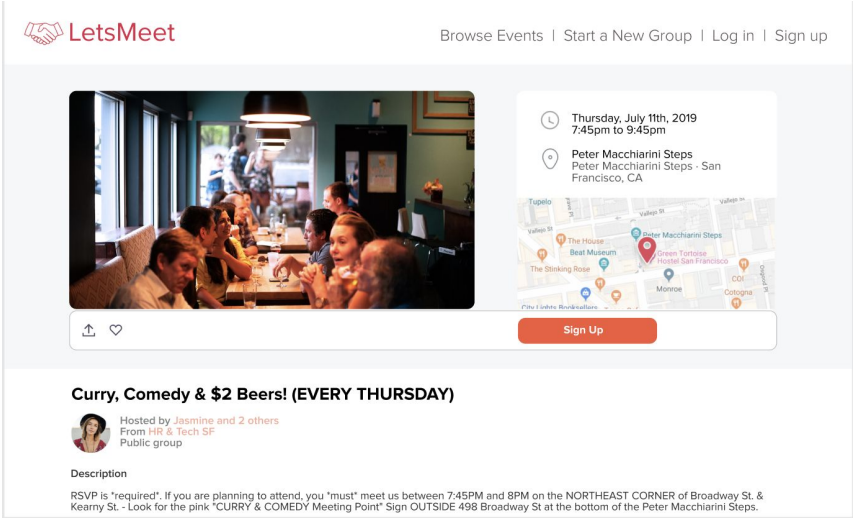


San Francisco

Why use LetsMeet?

People who use our website find new groups of friends in their city and make lasting connections.





Tech Team

You are the first data analyst on the tech team at LetsMeet. The tech team primarily consists of engineers, web designers, and app developers. LetsMeet hopes to include more data analysts and a data scientist as the company grows.

You will be managed by the VP of Engineering, Isabella. The following slides tell you a little about the people you will be working with.



Isabella

Position: VP of Engineering

Work Expertise: I have been in engineering for 8 years now, previously working with Apple, Google, and Amazon.

Fun Fact: I have published three books!

Hobbies: Scuba diving and dancing

Favorite Movie: *Back to the Future*

Myers-Briggs Type Indicator: ENFP



Quinn

Position: Engineer

Work Expertise: I taught myself how to code in high school, then used a bootcamp program to gain more engineering skills. This is my first full time position as an Engineer.

Fun Fact: I once swam with the dolphins in Hawaii.

Hobbies: Collecting sunglasses, tasting coffee in every state I visit.

Favorite Movie: *Spirited Away*

Myers-Briggs Type Indicator: INTJ



Cat

Position: Director of Sales

Work Expertise: After starting my own water color business, I worked in sales for the first time for my own company. After realizing how much I enjoyed it, I started pursuing sales positions at other companies. Before joining the LetsMeet team, I was the Director of Sales at REI.

Fun Fact: In my spare time, I teach watercolor painting classes.

Hobbies: Hiking, swimming, biking, boating, going to the beach.

Favorite Movie: *The Wizard of Oz*

Myers-Briggs Type Indicator: ESFP



Evelyn

Position: CEO

Fun Fact: I have run three marathons!

Work Expertise: After working as a business analyst with Lululemon, Apple, and LinkedIn, I have taken on the role of CEO at LetsMeet.

Hobbies: Meditation and running.

Favorite Movie: *The Secret Life of Walter Mitty*

Myers-Briggs Type Indicator: ESTP

The Data

Our data exists in a .sql file. Although we have looked at the data, you will be the first person to analyze the data.

The following slides provide an overview of our database, including details of each table.

Tables in the Database

1. **grp**
   Groups that make up LetsMeet
2. **grp_member**
   Group members that make up groups
3. **venue_**
   Where groups meet
4. **event**
   Events that groups put on
5. **category**
   Different categories of groups
6. **city**
   Information about cities that host groups

grp

The core part of LetsMeet are the groups. The grp table contains information about the groups that use LetsMeet to network. Our groups cover a huge range of topics such as animal rights, backgammon, kayaking, and dachshund.

Columns: group_id, category_id, city_id, created, description, join_mode, group_name, rating, visibility, members, who.

Notes - There are more groups than we can keep track of. That's why we need you!

group_id	category_id	city_id	created	description	join_mode	group_name
6388	14	10001	2002-11-21 16:50:46	Those who practice or hold a strong interest in t…	open	Alternative Health NYC
6510	4	10001	2003-05-20 14:48:54	The New York Alternative Energy Meetupis for t…	open	Alternative Energy Meetup
8458	26	10001	2004-03-27 09:55:41	not_found	open	NYC Animal Rights
10359	16	10001	2003-05-22 14:19:52	Hello! Michael Kim 'Jamal' Riegelman, the Dire…	approval	NYC International Arabic Language & Culture Club
12111	16	10001	2002-10-08 17:22:28	NYC ASL is a social/conversation group for Dea…	open	The New York City American Sign Language Meetup Group
15217	31	10001	2003-02-05 02:32:46	This is a freewheeling friendly group of terrific p…	open	The New York City Backgammon Meetup Group

grp_member

The grp_member table contains data on individual members. Each row lists one member and one group. So members in multiple groups appear multiple times. Most members are based in the New York, Chicago, and San Francisco areas, but they come from all over the world.

Columns: member_id, city, country, hometown, joined, member_name, member_status, visited, group_id.

Notes - Members do not belong to LetsMeet. Members belong to groups. Technically groups do not belong to LetsMeet either. Membership comes from LetsMeet networking.

member_id	city	country	hometown	joined	member_name	member_status	visited	group_id
227	San Francisco	us	not_found	2016-02-20 03:17:19	Patrick	active	2016-02-26 19:34:55	19335148
848	New York	us	not_found	2015-07-13 22:49:41	alex	active	2016-06-28 20:41:22	8841422
3811	San Francisco	us	not_found	2016-01-08 19:32:37	David	active	2016-01-26 19:35:01	19253477
6688	Chicago	us	not_found	2008-04-09 20:36:10	Matt	active	2014-05-25 19:30:47	578609
9098	New York	us	not_found	2009-02-13 16:14:24	Michael	active	2009-10-26 19:44:01	287576
9375	New York	us	NYC / Melbourne	2016-09-21 21:06:35	Ali	active	2017-06-12 21:26:47	20397238

event

Events are what the members gather for. This table contains the most columns. Included are text descriptions, waitlist information, rsvp information, urls, and more.

Columns: event_id, created, description, duration, event_url, group_id, headcount, how_to_find_us, maybe_rsvp_count, event_name, photo_url, rsvp_limit, event_status, event time, updated, utc_offset, venue_id, visibility, waitlist_count, why, yes_rsvp_count.

Notes - Some of these columns may be extraneous.

event_id	created	description	duration	event_url	group_id	headcou
153868222	2013-12-03 21:24:29	Solve the murder case while eating a 3 course…	20700	https://www.letzmeet.com/SanFranciscoStartupFun/event…	5817262	0
184167702	2014-05-20 18:52:00	NOTE: This event is EVERY FRIDAY!! Signup i…	77400	https://www.letzmeet.com/hr-and-tech-sf/events/184167702/	1627081	0
215200502	2014-10-23 16:18:44	We'll be heading out with a group of travelers fo…	37800	https://www.letzmeet.com/hr-and-tech-sf/events/215200502/	1627081	0
220826782	2015-02-28 19:27:32	NOTE: This event is EVERY FRIDAY!! Signup i…	113400	https://www.letzmeet.com/SanFranciscoStartupFun/event…	5817262	0
227948102	2016-01-08 21:35:40	Aussie Aussie Aussie, Oi Oi Oi! Join us for the F…	36000	https://www.letzmeet.com/hr-and-tech-sf/events/227948102/	1627081	0
227967930	2016-01-09 20:41:06	Aussie Aussie Aussie, Oi Oi Oi! Join us for the F…	36000	https://www.letzmeet.com/SanFranciscoStartupFun/event…	5817262	0

venue_

This table contains data on the venues that host LetsMeet events. Standard information like latitude and longitude along with the address are provided.

Columns: venue_id, address_1, address_2, city, country, distance, lat, localized_country_name, lon, venue_name, phone, state, zip, RAND() DOUBLE.

Notes - We often get requests about venues from group organizers.

venue_id	address_1	address_2	city	country	distance	lat	localized_country_name	lon	venue_name
11898	7000 North Glenwood		Chicago	us	0.00	42.00999800	USA	-87.66999800	Heartland Cafe
16753	105 1st Ave.		New York	us	0.00	40.72683300	USA	-73.98633600	Counter Vegetarian Re
96548	58 Avenue B (corner of E. 4th St.)		New York	us	0.00	40.72328200	USA	-73.98251300	Kate's Joint
114587	550 Madison Avenue on the west side		New York	us	0.00	40.71429800	USA	-74.00669900	Sony Public Arcade
125034	126 East 28th Street		New York	us	0.00	40.74275300	USA	-73.98301800	Copper Chimney Resta
142528	26 W. Hubbard St.		Chicago	us	0.00	41.89006800	USA	-87.62908900	Howl at the Moon

category

We organize our groups into the following categories: Arts & Culture, Career & Business, Cars & Motorcycles, Community & Environment, Dancing, Hobbies & Crafts, Language & Ethnic Identity, Lifestyle, Book Clubs, Movies & Film, Music, New Age & Spirituality, Outdoor & Adventure, Paranormal, Parents & Family, Pets & Animals, Photography, Religion & Beliefs, Sci-Fi & Fantasy, Singles, Socializing, Sports and Recreation, Support, Tech, Writing.

Columns: category_id, category_name, shortname, sort_name.

This table is the shortest and easiest to make sense of.

category_id	category_name	shortname	sort_name
1	Arts & Culture	Arts	Arts & Culture
2	Career & Business	Business	Career & Business
3	Cars & Motorcycles	Auto	Cars & Motorcycles
4	Community & Environment	Community	Community & Environment
5	Dancing	Dancing	Dancing
6	Education & Learning	Education	Education & Learning

city

This table contains information about the cities where LetsMeet groups and members gather. In addition to state, zip, country, and latitude/longitude, a column member_count displays the number of members per city.

Columns: city_id, city, state, zip, country, localized_country_name, distance, longitude, latitude, member_count, ranking.

Notes - The city table could use a little cleaning.

city_id	city	state	zip	country	localized_country_name	distance	longitude	latitude	member_count	ranking
7093	West New York	NJ	7093	us	USA	2524.541	-74.01000214	40.79000092	661	32
10001	New York	NY	10001	us	USA	2526.837	-73.98999786	40.75000000	229371	0
13417	New York Mills	NY	13417	us	USA	2392.162	-75.29000092	43.09999847	22	109
46312	East Chicago	IN	46312	us	USA	1810.371	-87.45999908	41.63999939	31	90
56567	New York Mills	MN	56567	us	USA	1418.834	-95.34999847	46.68999863	5	1
60064	North Chicago	IL	60064	us	USA	1779.682	-87.86000061	42.33000183	45	32

Your Role

We're excited to finally have someone who can analyze the data to gain new insights and drive business decisions. Since we are growing rapidly, this role will become increasingly important. Your role will include the following:

- Respond to sales requests
- Respond to engineering requests
- Respond to Evelyn (CEO) requests
- Analyze data for insights
- Improve database structure
- Write reports to summarize data

All requests will come from questions that we need a data analyst to answer!

Welcome to the team!

LetsMeet